

SANDHYA PREMNATH TIWARI

Independent Associate Professor & Principal Investigator

Laboratory for Biomolecular Modeling and Dynamics | Institute for Protein Research, The University of Osaka
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RESEARCH PROFILE

Computational biophysicist investigating how protein structure and conformational dynamics shape molecular recognition, regulation, and function. My research integrates molecular dynamics simulations, elastic network and normal mode approaches, structural bioinformatics, integrative structural modeling, and machine learning. A central theme is understanding how collective motions are conserved or altered by mutations, ligand binding, oligomerization, and protein-protein interactions, and translating these insights into scalable computational approaches for biomolecular research.

ACADEMIC APPOINTMENTS

Independent Associate Professor & Principal Investigator | Oct 2022-Present

Institute for Protein Research, The University of Osaka, Japan

Lead the Laboratory for Biomolecular Modeling and Dynamics within the Advanced Data Science Center for Protein Research (ASPiRE). Direct an independent research program spanning protein dynamics, molecular simulation, integrative structural biology, cryo-EM-guided modeling, and machine-learning approaches; supervise graduate and visiting researchers and develop international collaborations.

Assistant Professor (Tenure Track; HIRAKU) | Graduate Mar 2021-Oct 2022

School of Integrated Sciences for Life, Hiroshima University, Japan

Faculty member in Molecular Biophysics. Developed an independent research program on protein dynamics, allostery, and protein-protein interactions; taught graduate-level courses.

RIKEN Special Postdoctoral Researcher | RIKEN Center for Apr 2018-Mar 2021

Computational Science, Kobe, Japan

Developed computational strategies for retrieving biomolecular 3D structures from sparse XFEL diffraction data and for improving single-particle cryo-EM reconstruction workflows.

Postdoctoral Researcher | RIKEN Advanced Institute for Mar 2016-Apr 2018

Computational Science, Kobe, Japan

Developed image-retrieval approaches for identifying candidate 3D structures from limited experimental cryo-EM projection images.

Researcher with Teaching Duties | University of Bergen, Nov 2014-Nov 2015

Norway

Studied conservation of intrinsic dynamics across homologous proteins and structural folds using elastic network models and comparative analysis; contributed to teaching and student supervision.

EDUCATION

PhD, Computational Biology | University of Bergen, Norway 2015

Thesis: "Computational Strategies for Comparing the Intrinsic Dynamics of Multiple Proteins."

MRes, Computational Biology | University of York, United Kingdom 2009

Thesis: "Characterising Protein-Protein Interactions within Transmembrane Oligomers"

BSc (Hons), Biological Sciences | Nanyang Technological University, Singapore 2007

SELECTED AWARDS & FELLOWSHIPS

- Exploration France 2023 Research Scholarship, Embassy of France in Japan - fully funded research exchange at I2BC, CNRS/CEA/Université Paris-Saclay (2023).
- RIKEN Special Postdoctoral Fellowship - competitive three-year fellowship with JPY 1 million annual research funding (2018-2021).
- Best Poster Award, International Society of Quantum Biology and Pharmacology President's Meeting, Bergen, Norway (2016).

SELECTED PUBLICATIONS

1. Bekker GJ, Tiwari SP. Uncovering the role of dimerization on dynamics and inhibitor stability in human versus *Xenopus* LTA4H. *Biophysical Journal*. 2026;125(16):4300-4312. doi:10.1016/j.bpj.2026.03.027.
2. Sudarsanam K, Srivastava A, Tiwari SP. Comparing PAS domain-coupled intrinsic dynamics in bHLH-PAS domain transcription factor complexes. *Biophysical Journal*. 2026;125(2):629-640. doi:10.1016/j.bpj.2025.10.038.
3. Dasgupta B, Tiwari SP. Explicit versus implicit consideration of binding partners in protein-protein complex to elucidate intrinsic dynamics. *Biophysical Reviews*. 2022;14(6):1379-1392. doi:10.1007/s12551-022-01026-5.
4. Tiwari SP, Tama F, Miyashita O. Protocol for retrieving three-dimensional biological shapes for a few XFEL single-particle diffraction patterns. *Journal of Chemical Information and Modeling*. 2021;61(8):4108-4119. doi:10.1021/acs.jcim.1c00602.

Full publication list: Google Scholar - scholar.google.no/citations?user=-UPKkMUAAAAJ

ResearchMap - researchmap.jp/sandhyatiwari

TEACHING, SUPERVISION & ACADEMIC LEADERSHIP

- Graduate teaching in protein structure and dynamics, molecular simulation, computational modeling, and structural bioinformatics.
- Supervision and mentoring of doctoral, master's, undergraduate, and visiting researchers in computational biomolecular science.
- Experience establishing independent research programs, developing international collaborations, and preparing competitive research proposals.

SELECTED RESEARCH TRAINING

Research Intern, European Bioinformatics Institute (EMBL-EBI), Hinxton, UK - under Prof. Dame Janet Thornton

May-Sep 2009

Research Intern, Bioinformatics Institute, A*STAR, Singapore - under Dr. Chandra Verma

Jun 2007-Jul 2008

PROFESSIONAL MEMBERSHIPS & LANGUAGES

Memberships: Biophysical Society of Japan; Protein Science Society of Japan; Biophysical Society (USA); The Protein Society (USA); American Association for the Advancement of Science (AAAS).

Languages: English (fluent); Hindi (fluent); Malay (basic); Norwegian (basic); Japanese (JLPT N5).

REFERENCES

Dr. Chandra Verma

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Prof. Florence Tama

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Prof. Nathalie Reuter

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Sandhya Tiwari is an Associate Professor and Principal Investigator of the Laboratory for Biomolecular Modeling and Dynamics at the Institute for Protein Research, The University of Osaka, Japan. Before joining The University of Osaka in 2022, she was an Assistant Professor at Hiroshima University and a Special Postdoctoral Researcher at RIKEN's Center for Computational Science. She received her PhD from the University of Bergen, Norway, where she developed computational strategies for comparing intrinsic protein dynamics. Her research focuses on understanding the relationship between protein structure, dynamics, and function using computational approaches, including molecular dynamics simulations, elastic network models, integrative structural modeling, and machine learning. She is particularly interested in how collective protein motions are conserved or altered by mutations, ligand binding, oligomerization, and molecular interactions.